

# Md. Forhan Shahriar Fahim

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## EDUCATION

### University of Rajshahi

Jan 2019 – Dec 2024

*B.Sc. in Computer Science & Engineering (CGPA: 3.66, Last 2 Years Avg: 3.82)*

*Rajshahi, Bangladesh*

- **Thesis:** Force Invariant EMG Pattern Recognition Using Deep Learning (Supervisor: Dr. Shamim Ahmad)
- **Relevant Coursework:** Data Structures (C++), Algorithms, OOP (Java), Database Systems, Operating Systems, AI, Digital Image Processing, Software Engineering, System Analysis.

## RESEARCH INTERESTS

AI Safety, AI Interpretability, Large Language Models, Computer Vision, Medical Imaging

## EXPERIENCE

### Pundra University of Science & Technology

Mar 2025 – Present

*Lecturer*

*Bogura, Bangladesh*

- Teach undergraduate courses in Design and Analysis of Algorithms and Machine Learning; have also taught Web Engineering, Structured Programming, and Simulation & Modeling.
- Design Outcome-Based Education (OBE)-aligned lectures, slides, learning activities, and assessments that connect course content with learning outcomes.
- Supervise undergraduate research theses in AI and Machine Learning, guiding students through problem formulation, experimental design, model evaluation, and manuscript preparation.

## PUBLICATIONS

- [1] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Nafiu Rahman, and Adrita Alam, “VulPatchNet: A Dual-Encoder Cross-Attention Framework for Multi-Language Vulnerability Patch Generation,” in *5th IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)* (Accepted).
- [2] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, and Md. Tanjid Sheikh, “Explainable Multi-Task Transformer-Based Framework for Cross-Language Source Code Vulnerability Detection with Auxiliary CWE and Language Classification,” *Journal of Cyber Security Technology*, vol. 10, no. 1, Art. no. 2700412, 2026. doi:10.1080/23742917.2026.2700412 [🔗](#).
- [3] Md. Tanjid Sheikh, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, Sumaya Akter Mahmuda, and S. M. Imran Reza, “Systematic Hybrid Analysis of Android Applications: CWE-Categorized Vulnerabilities and Practical Mitigation Strategies,” in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, pp. 1–6, 2026. doi:10.1109/QPAIN69676.2026.11546396 [🔗](#).
- [4] S. M. Imran Reza, Iffath Tanjim Moon, **Md. Forhan Shahriar Fahim**, Adrita Alam, and Md. Tanjid Sheikh, “An Empirical Analysis of Transformer-Based Models with LIME Explainability for JavaScript Vulnerability Detection,” in *2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, pp. 1–6, 2026. doi:10.1109/ICECTE69292.2026.11429428 [🔗](#).

## RESEARCH EXPERIENCE

### Systematic Review: SSL in Cancer Imaging | Lead Researcher

Aug 2025 – Sep 2026

- Reviewed Self-Supervised Learning (SSL) methods across MRI, CT, and histopathology using a PRISMA-guided approach to study label-efficient medical image analysis.
- Synthesized approaches, datasets, evaluation strategies, and research gaps to inform a manuscript for peer-reviewed publication.

## Force-Invariant EMG Pattern Recognition Using Deep Learning | Thesis

Apr 2023 – Sep 2024

- Developed a hybrid CNN–LSTM framework for force-invariant sEMG pattern recognition, benchmarking its performance against ResNet-50 and VGG-16 under varying contraction forces.
- Achieved 80% classification accuracy across dynamic contraction levels, demonstrating robustness to variations in muscle contraction force.

## TECHNICAL PROJECTS

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### CSE Academic Operations Hub | Next.js, TypeScript, PostgreSQL

Aug 2026 – Sep 2026

- Built a portal for scheduling, attendance, and faculty workload across HSC and Diploma cohorts.
- Blocked conflicting routines from publication with minute-level teacher, room, and cohort checks.
- Added CSV roster import, attendance summaries, extra-class payment sheets, and versioned exports.

### BookHive: Book Store Management | PHP, Laravel, MySQL, Bootstrap

Mar 2023 – Dec 2023

- Built a Laravel/MySQL application to manage book records and stock levels.
- Added catalog search and filtering through a responsive Bootstrap interface.

## ACADEMIC SUPERVISION

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- **Undergraduate Research Supervisor (10+ students):** Supervise undergraduate research in medical imaging, including *Brain Tumor Segmentation using Attention Mechanisms* and *Automated Tumor Detection*.
- **Capstone Project Supervisor:** Supervise full-stack development projects, including an *Artisanal E-commerce Platform* and a *Web-based Hostel Management System*, with emphasis on software development life cycle (SDLC), system design, implementation, and agile documentation.

## PROBLEM SOLVING

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### Competitive Programming | Codeforces, LeetCode, AtCoder

- Solved 1,000+ algorithmic problems across various platforms; achieved **Max Rating 1396 (Pupil)** on Codeforces and **1661** on LeetCode.

## HONORS & AWARDS

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- **Young Enthusiastic Faculty Member Award (2025), Pundra University of Science & Technology:** Recognized for teaching, student mentorship, and departmental service.
- **Dean's Award (2023), University of Rajshahi:** Recognized for outstanding undergraduate academic performance.

## TECHNICAL SKILLS

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- **Programming:** Python, C/C++, Java.
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face.
- **Software & Databases:** Next.js, Node.js, PostgreSQL, MySQL.
- **Tools:** Git, Linux, LaTeX.

## LEADERSHIP & CO-CURRICULAR ACTIVITIES

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- **Event Coordinator, PUB CSE Fest (2026):** Coordinated technical events and supported broader festival planning, sports logistics, and media coverage.
- **Competitive Programming Mentor:** Mentored students through the PUB and RU programming communities, conducting workshops on algorithms, problem solving, and national programming contest preparation.

## REFERENCES

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### Dr. Shamim Ahmad

Professor, Dept. of CSE,  
University of Rajshahi

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*Undergraduate thesis supervisor*

### Dr. Md. Ekramul Hamid

Professor, Dept. of CSE,  
University of Rajshahi

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